

ITZEHOE, Germany

WMO: 101420

Lat: 53.9897N

Lon: 9.5697E

Elev: 26

StdP: 101.02

Time Zone: 1.00 (EUC)

Period: 95-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.6	(c) -7.3	(d) -11.8	(e) 1.4	(f) -8.3	(g) -9.5	(h) 1.7	(i) -6.4	(j) 12.8	(k) 7.8	(l) 11.4	(m) 6.9	(n) 2.1	(o) 120	(p) 0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	28.4	18.7	26.1	18.3	24.1	17.7	20.3	25.7	19.3	24.2	18.3	22.8	3.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.4	22.4	17.5	12.5	21.3	16.5	11.8	20.4	58.3	25.8	55.1	24.3	51.9	23.2	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.4	(b) 8.1	(c) 7.1	(e) -13.7	(f) 32.5	(g) 3.5	(h) 1.7	(i) -16.3	(j) 33.7	(k) -18.3	(l) 34.7	(m) -20.3	(n) 35.6	(o) -22.9	(p) 36.8		
(4) 9.4	(5) 8.1	(6) 7.1	(7) -13.6	(8) 22.6	(9) 3.6	(10) 0.8	(11) -16.2	(12) 23.2	(13) -18.4	(14) 23.7	(15) -20.4	(16) 24.2	(17) -23.0	(18) 24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	1.4	1.8	3.7	8.4	11.9	14.6	17.3	17.3	14.0	9.7	5.3	1.7
	DBStd	6.88	4.45	4.28	3.98	3.81	3.31	3.18	3.06	2.90	2.93	3.46	3.70	4.39
	HDD10.0	1240	268	229	195	76	17	1	0	0	2	48	146	257
	HDD18.3	3486	526	462	452	299	201	121	58	57	134	268	392	516
	CDD10.0	858	0	1	2	27	77	138	226	225	122	38	3	0
	CDD18.3	62	0	0	0	0	2	8	26	23	3	0	0	0
	CDH23.3	600	0	0	0	5	27	94	250	197	26	1	0	0
	CDH26.7	148	0	0	0	0	4	23	67	53	2	0	0	0
Wind	WSAvg	3.2	3.7	4.0	3.8	3.2	3.1	2.9	2.8	2.6	2.7	2.9	3.1	3.2
Precipitation	PrecAvg	861	72	60	55	41	61	81	92	88	79	71	75	84
	PrecMax	1103	160	148	123	93	141	151	232	227	191	218	156	193
	PrecMin	606	2	11	7	3	13	4	32	27	19	30	11	26
	PrecStd	125	37	31	29	21	27	36	51	43	42	37	40	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	12.0	16.3	23.7	26.4	29.5	31.7	30.7	26.2	19.7	14.5	11.7
		MCWB	9.7	9.0	10.9	14.7	16.9	18.8	20.3	20.0	18.4	15.8	12.4	10.3
	2%	DB	9.9	10.2	13.1	20.4	23.3	26.1	28.5	28.2	23.4	17.3	12.7	9.9
		MCWB	8.4	8.2	9.5	13.3	15.4	18.0	19.1	18.7	17.6	14.6	11.2	8.9
	5%	DB	8.5	8.8	11.1	17.8	21.1	23.3	26.2	25.4	20.8	15.9	11.5	8.6
		MCWB	7.4	7.3	8.6	11.8	14.3	16.9	18.6	18.4	16.2	13.5	10.1	7.5
	10%	DB	7.2	7.5	9.4	15.1	18.8	20.8	23.9	23.0	18.9	14.7	10.4	7.5
		MCWB	6.3	6.2	7.3	10.6	13.4	15.6	18.0	17.7	15.0	12.8	9.1	6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	10.3	11.8	15.3	18.6	20.8	21.7	21.9	19.5	16.3	12.9	10.5
		MCDB	11.3	11.4	14.1	21.8	24.0	26.9	27.1	28.3	23.7	18.9	14.0	11.5
	2%	WB	8.5	8.5	10.2	13.6	16.7	19.1	20.4	20.0	18.1	14.9	11.6	8.8
		MCDB	9.3	9.8	12.6	19.2	21.4	24.3	26.2	24.9	22.6	16.8	12.6	9.7
	5%	WB	7.5	7.3	8.8	12.4	15.0	17.4	19.4	19.1	16.7	13.9	10.4	7.6
		MCDB	8.3	8.6	10.9	17.1	19.3	21.5	24.6	23.8	19.9	15.5	11.3	8.4
	10%	WB	6.3	6.2	7.5	11.0	13.8	16.2	18.3	18.1	15.7	13.0	9.3	6.7
		MCDB	7.3	7.4	9.2	14.5	17.9	20.0	22.8	22.2	18.5	14.4	10.2	7.4
Mean Daily Temperature Range	5% DB	MDBR	4.3	5.0	6.9	9.5	10.1	9.3	9.5	9.2	8.4	7.0	5.0	4.5
		MCDBR	4.7	5.2	9.7	13.9	14.3	13.9	14.0	13.1	11.4	7.8	5.0	4.6
		MCWBR	4.2	4.1	6.8	7.7	8.1	7.2	6.4	6.2	6.6	5.3	4.1	4.2
	5% WB	MCDBR	4.6	4.8	8.6	12.1	12.0	11.6	12.1	11.1	10.0	7.1	4.7	4.6
		MCWBR	4.3	4.0	6.4	7.1	7.5	6.8	6.1	6.0	6.0	5.2	4.2	4.3
Clear-Sky Solar Irradiance	taub	0.320	0.338	0.367	0.412	0.404	0.402	0.420	0.414	0.388	0.356	0.334	0.311	
	taud	2.318	2.329	2.277	2.195	2.260	2.291	2.258	2.286	2.342	2.409	2.363	2.312	
	Ebn at Noon	630	748	804	812	842	847	822	803	775	719	609	560	
	Edh at Noon	59	79	103	127	126	123	126	115	97	73	56	49	
All-Sky Solar Radiation	RadAvg	0.54	1.20	2.43	4.05	5.10	5.31	5.03	4.13	2.97	1.62	0.68	0.40	
	RadStd	0.06	0.17	0.23	0.47	0.54	0.43	0.63	0.31	0.34	0.18	0.07	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-142	N/A	N/A

Nomenclature: See separate page